

WeBWorK assignment number WW_03 is due : 07/28/2008 at 11:00pm CDT.

The

(* replace with url for the course home page *)

for the course contains the syllabus, grading policy and other information.

This file is /conf/snippets/setHeader.pg you can use it as a model for creating files which introduce each problem set.

The primary purpose of WeBWorK is to let you know that you are getting the correct answer or to alert you if you are making some kind of mistake. Usually you can attempt a problem as many times as you want before the due date. However, if you are having trouble figuring out your error, you should consult the book, or ask a fellow student, one of the TA's or your professor for help. Don't spend a lot of time guessing – it's not very efficient or effective.

Give 4 or 5 significant digits for (floating point) numerical answers. For most problems when entering numerical answers, you can if you wish enter elementary expressions such as $2 \wedge 3$ instead of 8, $\sin(3 * \pi/2)$ instead of -1, $e \wedge (\ln(2))$ instead of 2, $(2 + \tan(3)) * (4 - \sin(5)) \wedge 6 - 7/8$ instead of 27620.3413, etc. Here's the [list of the functions](#) which WeBWorK understands.

You can use the Feedback button on each problem page to send e-mail to the professors.

1. (1 pt) 2360lib/chap3/leon.3.4/n01.pg

(S. 3-4, cf. no. 1)

Determine whether the following vectors form a basis for R^2 :

(a) $\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

- yes
- no

(b) $\begin{pmatrix} 3 \\ 2 \end{pmatrix}, \begin{pmatrix} -6 \\ -4 \end{pmatrix}$

- yes
- no

(c) $\begin{pmatrix} 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 7 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

- yes
- no

(d) $\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

- yes
- no

2. (1 pt) 2360lib/chap3/leon.3.4/n02.pg

(S. 3-4, cf. no. 2)

Determine whether the following vectors form a basis for R^3 :

(a) $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

- yes
- no

(b) $\begin{pmatrix} 3 \\ 2 \\ 7 \end{pmatrix}, \begin{pmatrix} 6 \\ 1 \\ -4 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

- yes

- no

(c) $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 5 \\ 0 \end{pmatrix}$

- yes
- no

(d) $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}, \begin{pmatrix} 4 \\ -7 \\ 3 \end{pmatrix}, \begin{pmatrix} 8 \\ 2 \\ -5 \end{pmatrix}$

- yes
- no

(e) $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}, \begin{pmatrix} 4 \\ -7 \\ 3 \end{pmatrix},$

- yes
- no

3. (1 pt) 2360lib/chap3/leon.3.4/n04.pg

(S. 3-4, cf. no. 4)

Given the vectors:

$$x_1 = \begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix}, \quad x_2 = \begin{pmatrix} 6 \\ 4 \\ 4 \end{pmatrix}, \quad x_3 = \begin{pmatrix} 0 \\ 16 \\ 12 \end{pmatrix}$$

What is the dimension of $\text{Span}(x_1, x_2, x_3)$?

4. (1 pt) 2360lib/chap3/leon.3.4/n07.pg

(S. 3-4, cf. no. 7)

Find a basis for the subspace S of R^4 consisting of all vectors of the form $(a - 2b + c, b - c, c, b + a)^T$, where a , b , and c range over all real numbers. (Enter your answer as a list of vectors separated by commas, and remember that a vector is entered using angular brackets $\langle$ and $\rangle$, e.g. $(1, 2, 3)^T$ would be entered as $\langle 1, 2, 3 \rangle$.)

Basis: _____

What is the dimension of S ? _____

5. (1 pt) 2360lib/chap3/leon.3.4/n11.pg

(S. 3-4,)

Find a basis for the null space $N(A)$ of the matrix $A =$

$$\begin{bmatrix} 1 & 3 & 2 & 1 \\ 3 & 1 & 1 & 2 \end{bmatrix}.$$

Basis: _____

(Enter your answers as vectors separated by commas.)

What is the dimension of $N(A)$? _____

6. (1 pt) 2360lib/chap3/leon.3.4/ur.la.10.1.pg

The vectors

$$v_1 = \begin{bmatrix} 1 \\ -4 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} -6 \\ 4 \\ -4 \end{bmatrix}, \text{ and } v_3 = \begin{bmatrix} -15 \\ 0 \\ k \end{bmatrix}$$

form a basis for $\mathbb{R}^3$ if and only if $k \neq$ ____.

7. (1 pt) 2360lib/chap3/leon.3.4/ur.la.10.28.pg

Find a basis of the row space of the matrix

$$A = \begin{bmatrix} 0 & -3 & -4 & 1 \\ -8 & 0 & 4 & 8 \\ -4 & 0 & 2 & 4 \end{bmatrix}.$$

$$\begin{bmatrix} _ \\ _ \\ _ \\ _ \end{bmatrix}, \begin{bmatrix} _ \\ _ \\ _ \\ _ \end{bmatrix}.$$

8. (1 pt) 2360lib/chap3/leon.3.4/ur.la.10.32.pg

Find a basis of the subspace of $\mathbb{R}^4$ consisting of all vectors of

$$\text{the form } \begin{bmatrix} x_1 \\ -8x_1 + x_2 \\ 5x_1 - 7x_2 \\ -5x_1 + 3x_2 \end{bmatrix}.$$

$$\begin{bmatrix} _ \\ _ \\ _ \\ _ \end{bmatrix}, \begin{bmatrix} _ \\ _ \\ _ \\ _ \end{bmatrix}.$$

9. (1 pt) 2360lib/chap3/leon.3.5/n01.pg

(S. 3-5, cf no 1,2)

For each of the following, find the transition matrix corresponding to the change in basis from $[\mathbf{u}_1, \mathbf{u}_2]$ to $[\mathbf{e}_1, \mathbf{e}_2]$.

(a) $\mathbf{u}_1 = (1, 1)^T, \mathbf{u}_2 = (1, 2)^T$

Transition matrix: _____

(b) $\mathbf{u}_1 = (2, 7)^T, \mathbf{u}_2 = (1, 4)^T$

Transition matrix: _____

(c) $\mathbf{u}_1 = (-1, 1)^T, \mathbf{u}_2 = (2, 2)^T$

Transition matrix: _____

For each of the ordered bases $[\mathbf{u}_1, \mathbf{u}_2]$ in the previous part of this problem, find the transition matrix corresponding to the change in basis from $[\mathbf{e}_1, \mathbf{e}_2]$ to $[\mathbf{u}_1, \mathbf{u}_2]$.

(a) _____

(b) _____

(c) _____

10. (1 pt) 2360lib/chap3/leon.3.5/n05.pg

(S. 3-5, cf no 5)

Let $\mathbf{u}_1 = (1, 2, 1)^T, \mathbf{u}_2 = (2, 5, 2)^T, \mathbf{u}_3 = (4, 8, 5)^T$.

(a) Find the transition matrix corresponding to the change of basis from $[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3]$ to $[\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3]$.

(b) Find the coordinate vector of each of the following vectors with respect to $[\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3]$. Remember to enter your response as a *vector* using angular brackets.

(i) $(2, 0, -3)^T$: _____

(ii) $(1, 1, 1)^T$: _____

(iii) $(0, 1, 0)^T$: _____

11. (1 pt) 2360lib/chap3/leon.3.5/n06.pg

(S. 3-5, cf no 6)

Let $\mathbf{v}_1 = (3, 1, 1)^T, \mathbf{v}_2 = (0, 1, 0)^T, \mathbf{v}_3 = (0, 1, 1)^T$, and let $\mathbf{u}_1 = (1, 2, 1)^T, \mathbf{u}_2 = (2, 5, 2)^T, \mathbf{u}_3 = (4, 8, 5)^T$.

(Note that $[\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3]$ is the basis from the previous exercise.)

(a) Find the transition matrix corresponding to the change of basis from $[\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3]$ to $[\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3]$.

(b) If $\mathbf{x} = 2\mathbf{v}_1 + 3\mathbf{v}_2 - 4\mathbf{v}_3$, express $\mathbf{x}$ as a linear combination of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$:

$$\mathbf{x} = _ \mathbf{u}_1 + _ \mathbf{u}_2 + _ \mathbf{u}_3.$$

12. (1 pt) 2360lib/chap3/leon.3.5/ur.la.10.3.pg

The set $B = \left\{ \begin{bmatrix} -4 \\ -6 \end{bmatrix}, \begin{bmatrix} 4 \\ 14 \end{bmatrix} \right\}$ is a basis for $\mathbb{R}^2$.

Find the coordinates of the vector $x = \begin{bmatrix} 12 \\ 34 \end{bmatrix}$ relative to the basis B :

$$[x]_B = \begin{bmatrix} _ \\ _ \end{bmatrix}$$

13. (1 pt) 2360lib/chap3/leon.3.5/ur.la.10.4.pg

Find the coordinate vector of $x = \begin{bmatrix} 3 \\ -3 \\ 0 \end{bmatrix}$ with respect to the

basis $B = \left\{ \begin{bmatrix} 1 \\ 7 \\ 7 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -4 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ or $\mathbb{R}^3$.

$$[x]_B = \begin{bmatrix} _ \\ _ \\ _ \end{bmatrix}$$

14. (1 pt) 2360lib/chap3/leon.3.6/n01.pg

(S. 3-6, cf no 1)

For each of the following matrices, find a basis for the row space, a basis for the column space, and a basis for the null space. (For column and null spaces, write your answers as a comma-separated list of column vectors using the angular bracket notation of WebWork. For row spaces, write your answers as a comma-separated list of 1xn (row) matrices.)

(a) $\begin{bmatrix} 1 & 5 & 7 \\ 2 & 3 & 1 \\ 1 & -16 & -32 \end{bmatrix}$

Basis for row space (set of row vectors): _____

Basis for column space (set of column vectors): _____

Basis for null space (set of column vectors): _____

(b) $\begin{bmatrix} 1 & 3 & 2 & 4 \\ 2 & 3 & 1 & 2 \\ -3 & 1 & -2 & 2 \end{bmatrix}$

Basis for row space (set of row vectors): _____

Basis for column space (set of column vectors): _____

Basis for null space (set of column vectors): _____

(c) $\begin{bmatrix} 1 & -3 & 3 & 4 \\ 2 & 1 & -1 & -2 \\ 8 & -3 & 4 & 2 \end{bmatrix}$

Basis for row space (set of row vectors): _____

Basis for column space (set of column vectors): _____

Basis for null space (set of column vectors): _____

15. (1 pt) 2360lib/chap3/leon.3.6/n02.pg

(S. 3-6, cf no 2)

Determine the dimension of the subspace of R^3 spanned by the given vectors:

(a) $\begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} -6 \\ -2 \\ 4 \end{bmatrix}, \begin{bmatrix} 6 \\ 2 \\ 3 \end{bmatrix}$. Dimension: ____

(b) $\begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} -6 \\ 2 \\ 4 \end{bmatrix}, \begin{bmatrix} 6 \\ 2 \\ 3 \end{bmatrix}$. Dimension: ____

(c) $\begin{bmatrix} 2 \\ -2 \\ 4 \end{bmatrix}, \begin{bmatrix} 4 \\ -2 \\ 6 \end{bmatrix}, \begin{bmatrix} -3 \\ 2 \\ -5 \end{bmatrix}, \begin{bmatrix} -3 \\ 3 \\ 6 \end{bmatrix}$. Dimension: ____

(Show the student hint after 1 attempts:)

Hint: The span of the vectors is the column space of the matrix whose columns are the given vectors. How do we find the dimension of the column space?

16. (1 pt) 2360lib/chap3/leon.3.6/n04.pg

(S. 3-6, cf no 4,5)

For each of the following choices of A and $\mathbf{b}$, determine if $\mathbf{b}$ is in the column space of A and state whether the system $A\mathbf{x} = \mathbf{b}$ is consistent.

(a) $A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$

$\mathbf{b}$ in column space of A ?

- yes
- no

$A\mathbf{x} = \mathbf{b}$ is consistent?

- yes
- no

(b) $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 1 & 2 \\ 1 & 1 & 2 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

$\mathbf{b}$ in column space of A ?

- yes
- no

$A\mathbf{x} = \mathbf{b}$ is consistent?

- yes
- no

(c) $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 1 & 2 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 5 \\ 10 \\ 5 \end{bmatrix}$

$\mathbf{b}$ in column space of A ?

- yes
- no

$A\mathbf{x} = \mathbf{b}$ is consistent?

- yes
- no

(d) $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 2 \\ 5 \\ 2 \end{bmatrix}$

$\mathbf{b}$ in column space of A ?

- yes
- no

$A\mathbf{x} = \mathbf{b}$ is consistent?

- yes
- no

(e) For each of the consistent systems in parts (a)-(d), determine which have infinitely many solutions. Enter your answer as a list of letters (lower case) separated by commas, e.g., b, c, d.

17. (1 pt) 2360lib/chap3/leon.3.6/n08.pg

(S. 3-6, cf no 8)

Let A and B be 7×6 matrices. If $\dim N(A) = 3$, what is the rank of A ?

If the rank of B is 4, what is the dimension of $N(B)$?

18. (1 pt) 2360lib/chap3/leon.3.6/n10.pg

(S. 3-6, cf no 10)

Let A be a 4×4 matrix with reduced row echelon form given by

$$\begin{bmatrix} 1 & 0 & 1 & 3 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

If

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ -3 \\ 2 \\ 1 \end{bmatrix}, \text{ and } \mathbf{a}_2 = \begin{bmatrix} 3 \\ -4 \\ 2 \\ 8 \end{bmatrix}$$

(remember $\mathbf{a}_i$ is the i th column of A) determine

$\mathbf{a}_3 =$ _____ (enter as a vector using angular brackets)

and

$\mathbf{a}_4 =$ _____,

so A is

19. (1 pt) 2360lib/chap3/leon.3.6/ur.la.10.8.pg

$$\text{Let } A = \begin{bmatrix} 2 & -8 & -2 \\ 1 & -4 & -1 \end{bmatrix}.$$

Find bases of the null space and range of A .

$$\text{null space: } \begin{bmatrix} _ \\ _ \\ _ \end{bmatrix}, \begin{bmatrix} _ \\ _ \\ _ \end{bmatrix}.$$

$$\text{range: } \begin{bmatrix} _ \\ _ \end{bmatrix}.$$

20. (1 pt) 2360lib/chap3/leon.3.6/ur.la.10.13a.pg

$$\text{Let } A = \begin{bmatrix} -6 & 9 & 6 & 9 \\ 2 & -3 & -2 & -3 \end{bmatrix}.$$

Find a basis of $\text{nullspace}(A)$.

$$\begin{bmatrix} _ \\ _ \\ _ \\ _ \end{bmatrix}, \begin{bmatrix} _ \\ _ \\ _ \\ _ \end{bmatrix}, \begin{bmatrix} _ \\ _ \\ _ \\ _ \end{bmatrix}.$$

21. (1 pt) 2360lib/chap3/leon.3.6/ur.la.10.14.pg

$$\text{Let } A = \begin{bmatrix} -2 & -4 & 9 \\ 4 & 1 & -4 \\ -6 & -5 & 13 \\ 2 & -3 & 5 \end{bmatrix}.$$

Find a basis of the range of A .

$$\begin{bmatrix} _ \\ _ \\ _ \\ _ \end{bmatrix}, \begin{bmatrix} _ \\ _ \\ _ \\ _ \end{bmatrix}.$$

22. (1 pt) 2360lib/chap3/leon.3.6/ur.la.10.30.pg

Find a basis of the column space of the matrix

$$A = \begin{bmatrix} 0 & 2 & -2 & 1 \\ -2 & -2 & 2 & 1 \\ 4 & 0 & 0 & -4 \end{bmatrix}.$$

$$\begin{bmatrix} _ \\ _ \\ _ \end{bmatrix}, \begin{bmatrix} _ \\ _ \\ _ \end{bmatrix}.$$

23. (1 pt) 2360lib/chap4/leon.4.1/n01.pg

(S. 4-1, cf no 1)

Let $L : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by

$$L(\mathbf{x}) = L \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 \\ -x_1 \end{pmatrix}$$

We want to show that L is a linear transformation by showing that

$$L(a\mathbf{x} + b\mathbf{y}) = aL(\mathbf{x}) + bL(\mathbf{y})$$

for all scalars a, b .

We'll first compute the left side of this equation in two steps (remembering to use WebWork's angular bracket notation (e.g., $\langle x_1, x_2 \rangle$) for vectors. Enter subscripted variables x_1 as $x1$.):

$$a\mathbf{x} + b\mathbf{y} = a \langle x_1, x_2 \rangle + b \langle y_1, y_2 \rangle = _$$

Now take L of the preceding vector to get:

$$L(a\mathbf{x} + b\mathbf{y}) = _$$

OK, now let's compute the right side of the equation.

$$\text{First, } aL(\mathbf{x}) = _$$

$$\text{and } bL(\mathbf{y}) = _.$$

$$\text{So } aL(\mathbf{x}) + bL(\mathbf{y}) = _$$

If your answers are correct, the two sides should be equal.

24. (1 pt) 2360lib/chap4/leon.4.1/n02.pg

(S. 4-1)

Let $L : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by

$$L(\mathbf{x}) = L \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 x_2 \\ -x_1 \end{pmatrix}$$

Show that L is *not* a linear transformation by finding vectors $\mathbf{x}$, and $\mathbf{y}$ such that $L(\mathbf{x} + \mathbf{y}) \neq L(\mathbf{x}) + L(\mathbf{y})$:

$$\mathbf{x} = \langle x_1, x_2 \rangle = _$$

$$\mathbf{y} = \langle y_1, y_2 \rangle = _$$

Prove your answer by calculating (for your choice of $\mathbf{x}, \mathbf{y}$):

$$L(\mathbf{x} + \mathbf{y}) = _$$

and

$$L(\mathbf{x}) + L(\mathbf{y}) = \underline{\hspace{2cm}}$$

Note: This problem has infinitely many correct answers.

25. (1 pt) 2360lib/chap4/leon.4.1/n04.pg

(S. 4-1, cf. no. 4)

Let $L : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation. If $L((1,2)^T) = (2,3)^T$, then $L((-3,-6)^T) = \underline{\hspace{2cm}}$

26. (1 pt) 2360lib/chap4/leon.4.1/n18.pg

(S. 4-1, cf. no 18)

Define $L : P_3 \rightarrow P_3$ by $L(p(x)) = p(0)x^2 + p'(0)$. It is easy to verify that L is a linear transformation (and you should do this.) $L(ax^2 + bx + c) = \underline{\hspace{2cm}}$

Find the kernel of L . The kernel of L is the set of all $ax^2 + bx + c$ satisfying

$$a = \underline{\hspace{2cm}}$$

$$b = \underline{\hspace{2cm}}$$

$$c = \underline{\hspace{2cm}}$$

(Your answer may contain free variables. Use d, e, f, ... for the free variables.)

27. (1 pt) 2360lib/chap4/leon.4.1/ur.la.14.6.pg

A linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ whose matrix is

$$\begin{bmatrix} 1 & -5 & -5 \\ -2 & 10 & 6+k \end{bmatrix}$$

is onto if and only if $k \neq \underline{\hspace{2cm}}$.

28. (1 pt) 2360lib/chap4/leon.4.1/ur.la.14.9.pg

Which of the following transformations are linear?

- A. $\begin{cases} y_1 = 9x_1 - 5x_2 + 7x_3 \\ y_2 = 10x_2 - 4x_3 \\ y_3 = -3x_1 - 6x_2 \end{cases}$
- B. $\begin{cases} y_1 = 4 \\ y_2 = 6 \\ y_3 = 3 \end{cases}$
- C. $\begin{cases} y_1 = 3x_2 \\ y_2 = -8x_3 \\ y_3 = -2x_1 \end{cases}$
- D. $\begin{cases} y_1 = 0 \\ y_2 = x_1x_2 \end{cases}$
- E. $\begin{cases} y_1 = 4x_1 + x_2 \\ y_2 = -x_1 \end{cases}$
- F. $\begin{cases} y_1 = x_1 + 10 \\ y_2 = x_2 \end{cases}$

29. (1 pt) 2360lib/chap4/leon.4.2/n01.pg

(S. 4-2, cf. no 1)

For each linear transformation L , find the standard matrix representation of L .

(a) $L : \mathbb{R}^2 \rightarrow \mathbb{R}^2$: $L(\mathbf{x}) = \begin{pmatrix} 2x_2 \\ x_1 - x_2 \end{pmatrix}$

$$A = \underline{\hspace{2cm}}$$

(b) $L : \mathbb{R}^2 \rightarrow \mathbb{R}^2$: $L(\mathbf{x}) = x_1 \mathbf{e}_2$

$$A = \underline{\hspace{2cm}}$$

(c) $L : \mathbb{R}^3 \rightarrow \mathbb{R}^3$: $L(\mathbf{x}) = \begin{pmatrix} x_2 \\ 2x_1 - x_3 \\ x_1 - x_2 + 3x_3 \end{pmatrix}$

$$A = \underline{\hspace{2cm}}$$

30. (1 pt) 2360lib/chap4/leon.4.2/n02.pg

(S. 4-2, cf. no 2)

For each linear transformation $L : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, find the standard matrix representation of L .

(a) $L((x_1, x_2, x_3)^T) = (x_1 + x_2 + x_3, x_3)^T$

$$A = \underline{\hspace{2cm}}$$

(b) $L((x_1, x_2, x_3)^T) = (x_2, x_1)^T$

$$A = \underline{\hspace{2cm}}$$

31. (1 pt) 2360lib/chap4/leon.4.2/n06.pg

(S. 4-2, cf. no 6)

Let

$$\mathbf{b}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{b}_3 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

and let L be the linear transformation from $\mathbb{R}^2$ into $\mathbb{R}^3$ defined by

$$L(\mathbf{x}) = (x_1 + 2x_2)\mathbf{b}_1 - x_1\mathbf{b}_2 - x_1\mathbf{b}_3$$

Find the matrix A representing L with respect to the bases $[\mathbf{e}_1, \mathbf{e}_2]$ and $[\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3]$.

$$A = \underline{\hspace{2cm}}$$

32. (1 pt) 2360lib/chap4/leon.4.2/n12.pg

(S. 4-2, cf. no. 12)

The linear operator L defined by

$$L(p(x)) = p'(0) + 2p'(x)$$

maps P_3 into P_2 . Find the matrix representation of L with respect to the ordered bases $[x^2, x, 1]$ and $[1 + x, 1 - x]$.

$$A = \underline{\hspace{2cm}}$$

Use your answer to find the coordinate vector of $L(p(x))$ with respect to the ordered basis $[1 + x, 1 - x]$.

$p(x) = -x^2 + 2x + 7$. Coordinate vector of $L(p(x))$ is

33. (1 pt) 2360lib/chap4/leon.4.2/ur.la.14.1.pg

If $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a linear transformation such that

$$T \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \quad T \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \\ 3 \end{bmatrix},$$

$$\text{and } T \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix},$$

$$\text{then } T \begin{bmatrix} 1 \\ -5 \\ 4 \end{bmatrix} = \begin{bmatrix} \underline{\hspace{1cm}} \\ \underline{\hspace{1cm}} \\ \underline{\hspace{1cm}} \end{bmatrix}.$$

34. (1 pt) 2360lib/chap4/leon.4.2/ur.la.14.2.pg

If $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation such that

$$T \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -4 \\ 1 \end{bmatrix} \text{ and } T \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix},$$

then the standard matrix of T is $A = \begin{bmatrix} ___ & ___ \\ ___ & ___ \end{bmatrix}$.

35. (1 pt) 2360lib/chap4/leon.4.2/ur.la.14.4.pg

If $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation such that

$$T \begin{bmatrix} 1 \\ 6 \end{bmatrix} = \begin{bmatrix} -12 \\ 21 \end{bmatrix} \text{ and } T \begin{bmatrix} 4 \\ -3 \end{bmatrix} = \begin{bmatrix} 33 \\ 3 \end{bmatrix},$$

then the standard matrix of T is $A = \begin{bmatrix} ______ & ______ \\ ______ & ______ \end{bmatrix}$.

36. (1 pt) 2360lib/chap4/leon.4.2/ur.la.14.10.pg

Consider a linear transformation T from $\mathbb{R}^3$ to $\mathbb{R}^2$ for which

$$T \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \quad T \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 7 \\ 6 \end{bmatrix},$$

$$\text{and } T \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}.$$

Find the matrix A of T .

$$A = \begin{bmatrix} ___ & ___ & ___ \\ ___ & ___ & ___ \end{bmatrix}.$$

37. (1 pt) 2360lib/chap4/leon.4.2/ur.la.14.28.pg

Find the matrix M of the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$\text{given by } T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -8x_1 - 9x_2 \\ 2x_1 + x_2 \end{bmatrix}.$$

$$M = \begin{bmatrix} ___ & ___ \\ ___ & ___ \end{bmatrix}.$$